

Predictive Modeling Using Machine Learning

Internship Project Report

Objective

The objective of this project is to develop a machine learning model that predicts outcomes using the Iris dataset. The project demonstrates the complete supervised learning workflow including loading data, preprocessing, training, testing, evaluating, and visualizing the results.

Dataset

The Iris dataset is one of the most popular datasets for machine learning. It contains 150 flower samples with four features: sepal length, sepal width, petal length, and petal width. The target variable contains three flower species: Setosa, Versicolor, and Virginica.

Tools and Libraries

Python, Pandas, Matplotlib, Seaborn, and Scikit-learn.

Methodology

Algorithm Used

Decision Tree Classifier was selected because it is simple, easy to understand, and effective for classification tasks.

Steps Followed

- Import the required Python libraries.
- Load the Iris dataset using scikit-learn.
- Split the dataset into 80% training data and 20% testing data.
- Create a Decision Tree Classifier model.
- Train the model using the training dataset.
- Predict the labels of the testing dataset.
- Calculate the model accuracy.
- Display the confusion matrix using Seaborn heatmap.

Results and Conclusion

Results

The trained model achieved an accuracy of 1.0 (100%) on the test dataset. The confusion matrix showed that all test samples were classified correctly.

Conclusion

This project successfully demonstrated predictive modeling using machine learning. The Decision Tree Classifier accurately classified the Iris flower dataset. The project provided practical experience in data loading, model training, prediction, accuracy evaluation, and visualization using a confusion matrix.

Future Scope

Future improvements include testing additional algorithms such as Random Forest, Support Vector Machine, and Logistic Regression, comparing their performance, and using larger real-world datasets.